Existence of Flips and Minimal Model for Varieties of General Type

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What is Minimal Model Program?

From beginning of Algebraic Geometry one of its main goals has been to classify Projective varieties over $\mathbb{C}$. Classifying Projective varieties isomorphically is nearly impossible. However, it can be observed that two birational varieties often share common features, so it is good enough to classify varieties up to birational equivalence. **Minimal Model Program** is one of the main tools for birational classification of algebraic varieties. It's goal is to find a good representative from each birational class.

- Let C be a projective curve, possibly singular.
- $\pi : \tilde{C} \to C$ be the normalization morphism.
- $\tilde{C}$ is the **Unique** smooth projective curve in birational class of C.
- Let S be a projective surface.
- By Hironaka's resolution of singularity, we can construct a smooth surface $\tilde{S}$ with a morphism $\pi : \tilde{S} \to S$.
- In fact, for any point $x \in \tilde{S}$, we can consider $\tilde{\tilde{S}} = Bl_x(\tilde{S})$ which is a smooth surface, birational to S but not isomorphic to $\tilde{S}$.
- Repeating this process we can produce **infinitely many non-isomorphic** smooth surfaces in the same birational class.

Castelnuovo's contraction theorem

- Around 1920 Castelnuovo proved the following

  If a smooth projective surface contains a (-1)-curve C (i.e., $C \cong \mathbb{P}^1$ and $C^2 = -1$), then there exists a birational morphism to a smooth surface S' with $\pi : S \to S'$ such that $\pi(C)$ is a point.

  This is called '**Castelnuovo's contraction theorem**'.
- Repeated application of this theorem to a smooth projective surface S yields a smooth surface S_{min} containing no (-1)-curve.
- Such an S_{min} is called a **minimal surface**.

Definitions and Examples

Let X be a projective variety.

- A cartier divisor D is called **nef** if $D.C \geq 0$ for all curves $C \subseteq X$.
- Any ample divisor is nef by Nakai-Moishezon criteria.
- A nef divisor may not be ample. Take a smooth curve C, consider $D = p + q - 2r$, it is nef but not ample.

Definitions and Examples (Continued...)

Let X be a smooth projective variety.

- The **canonical sheaf** is defined as $\omega_X := \wedge^{dim X}\Omega^1_X$. The **canonical divisor** K_X is the divisor corresponding to the canonical sheaf.

Let X be a projective variety.

- A variety X is called called '**Fano Variety**' if $-K_X$ is ample.
- $\mathbb{P}^n$ is fano variety for all $n \geq 1$. In general, a hypersurface H_d of degree d is fano iff $d < n+1$.
  (By adjunction formula, $K_{H_d} = \mathcal{O}(d-n-1)|_{H_d}$).
- A variety X is called **Mori fiber space** if there exists a morphism $\pi : X \to Y$ with dim $X >$ dim Y, with fibers being Fano varieties.

Mori's Idea of MMP

- In higher dimensions(≥ 3), one of the main difficulties is the absence of a good contraction theorem analogous to Castelnuovo's contraction theorem for surfaces.
- Nearly six decades later, around 1980, Mori made the following remarkable observation : On a smooth projective surface, let C be a (-1) curve. Then by adjunction formula, $K_S.C = -1$.i.e K_S is not nef. Mori's idea was to replace the question

  Is there a (-1) curve on S ?

  by

  Is the canonical divisor K_X nef?

  A variety X is called **Minimal Model** if K_X is nef.

Contraction Theorem

Let X be a smooth projective variety.

Theorem

If K_X is not nef , then there exists a contraction $f : X \to Z$ such that $f_*(\mathcal{O}_X) = \mathcal{O}_Z$ and all curves contracted by f are K_X - negative.

Remark: Even if we start with a smooth variety, after contraction output may become singular. Hence instead of taking smooth variety, we consider normal projective variety with mild singularity.

Types of contraction

Let X be a normal projective variety with mild singularity such that K_X is not nef.
Let $f : X \to Z$ be a contraction.

- If $\dim X > \dim Z$ then f is a Mori fiber space contraction.
- If f is birational, we have two cases.
  1. If $\mathrm{Exp}(f)$ contains a divisor, we call f to be a **Divisorial Contraction**.
  2. If $\mathrm{Exp}(f)$ has codimension at least 2, we call f to be **flipping contraction**.

Let $Div(X)$ be the Cartier divisor class group.

- We say D_1 is numerically equivalent to D_2 if $D_1.C = D_2.C$ for all curves $C \subset X$.
- $N^1(X) = Div(X)/\equiv$ is a free abelian group of finite rank.The group is called **Neron-Severi Group**. The rank is called Picard number of X.

Let $Z_1(X)$ be free abelian group generated by all curves on X.

- We say C_1 is equivalent to C_2 if $D.C_1 = D.C_2$ for all divisors D on X.
- We define $N_1(X) = Z_1(X)/\sim$

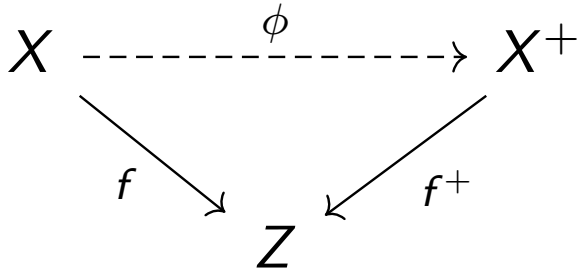
Tensoring with $\mathbb{R}$, we get $N^1(X)$ and $N_1(X)$ are dual vector spaces of each other.

Remark : If $f : X \to Z$ be a contraction. Then $\rho(Z) = \rho(X) - 1$.

Flip

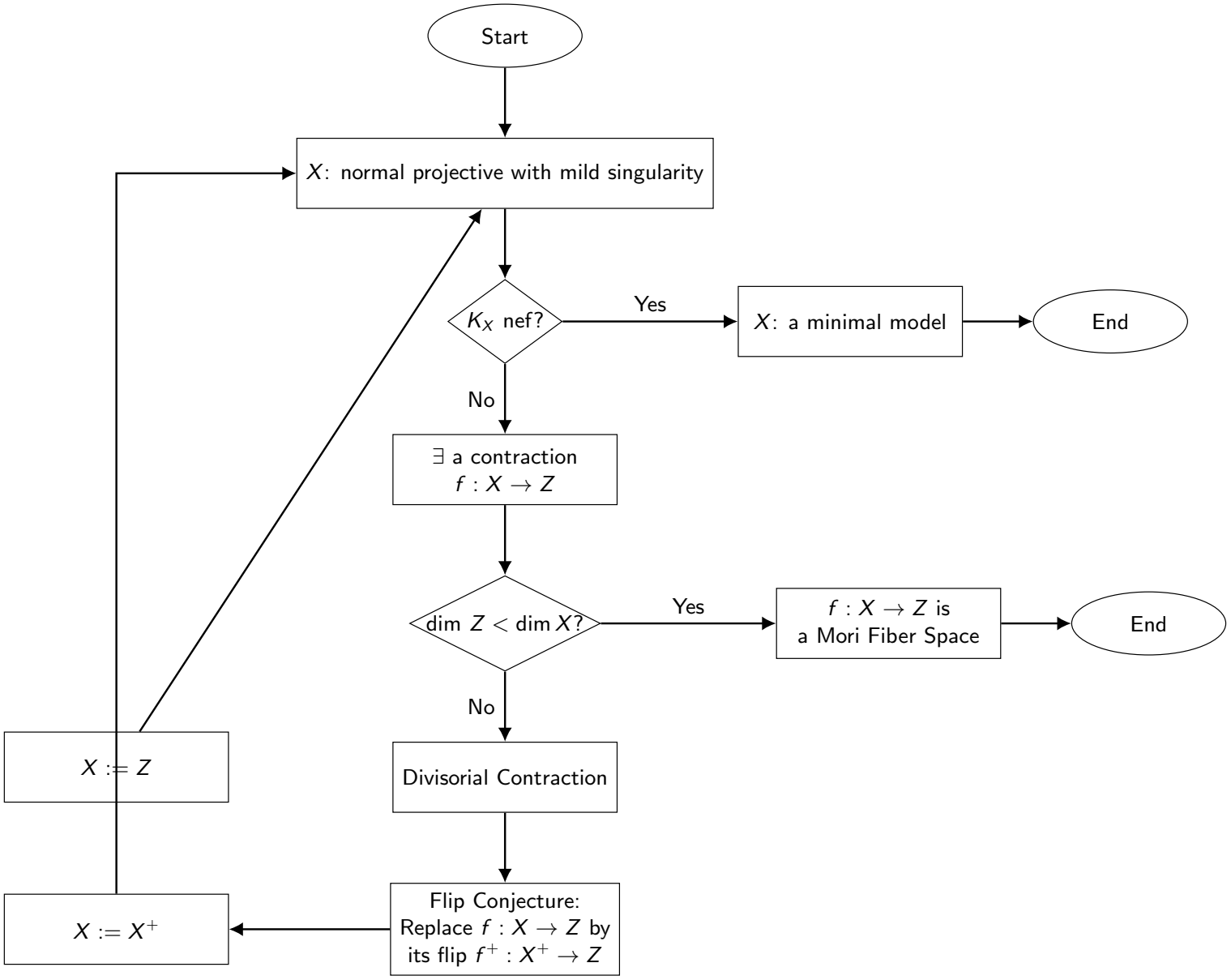
- Note that, If f is flipping contraction, then K_Z is not Cartier. So we want to replace Z by a new projective variety X^+.

  Definition(Flip): Let $f : X \to Z$ be a flipping contraction. We say $f^+ : X^+ \to Z$ is flip of f if f^+ is small birational morphism with K_{X^+} being f^+ ample.



- The picard rank $\rho(X) = \rho(X^+)$.

Algorithm of Minimal Model Program



- Since each divisorial contraction decreases the Picard number by 1, there can only be finitely many such contractions.
- For flips, the Picard number remains unchanged.

We need to establish two points:

1. The existence of flips.
2. The termination of flips.

Canonical ring

Definition: Let X be a normal projective variety with mild singularity with K_X being canonical divisor on X.

$$R(K_X) = \bigoplus_{m \geq 0} H^0(X, \mathcal{O}_X(mK_X))$$

$R(K_X)$ is called canonical ring of X.

Theorem

The canonical ring is preserved throughout each step of Minimal Model Program.

Proof:

- Let $f : X \to Z$ be a divisorial contraction. We can write

$$K_X = f^* K_Z + E$$

By negativity lemma we get $E \geq 0$.

$$H^0(X, \mathcal{O}_X(mK_X)) \cong H^0(Z, f_* \mathcal{O}_X(mK_X))$$

$$\cong H^0(Z, f_* \mathcal{O}_X(mE) \otimes \mathcal{O}_Z(mK_Z)) \cong H^0(Z, \mathcal{O}_Z(mK_Z))$$

- For flipping contraction we get

$$H^0(X, \mathcal{O}_X(mK_X)) \cong H^0(Z, f_* \mathcal{O}_X(mK_X)) \cong H^0(Z, \mathcal{O}_Z(mK_Z)))$$

$$H^0(X^+, \mathcal{O}_X(mK_{X^+})) \cong H^0(Z, f_*^+ \mathcal{O}_{X^+}(mK_{X^+})) \cong H^0(Z, \mathcal{O}_Z(mK_Z))$$

When does flip exist?

Theorem

Let X be a normal projective variety with mild singularity. $f : X \to Z$ be a contraction such that f is small. Then f^+ (flip) exists if and only if the $\bigoplus_{m\geq 0} f_*\mathcal{O}_X(mK_X)$ is a finitely generated $\mathcal{O}_Z$ algebra.

Proof.

Assume Z is affine. Then we have

$$H^0(X, \mathcal{O}_X(mK_X)) \cong H^0(X^+, \mathcal{O}_X(mK_{X^+}))$$

Then $R(K_X)$ is finitely generated.
Conversely ,$X^+ := Proj \bigoplus_{m\geq 0} f_*\mathcal{O}_X(mK_X)$. The map $f^+ : X^+ \to Z$ is isomorphism over $Z \setminus f(Expf)$. Now by using the fact $\mathcal{O}_Z(mK_Z)$ is a reflexive sheaf, we can say f^+ is small. □

Shokurov's reduction to pl flip contraction

Definition : Let (X, Δ) be a pair with mild singularity. Let $\Delta = \sum a_i D_i$ such that $a_i \in [0,1] \cap \mathbb{Q}$. We define $\lfloor \Delta \rfloor = \lfloor a_i \rfloor D_i$.$f : X \to Y$ be a flipping contraction such that $\lfloor \Delta \rfloor = S$ irreducible and $-S$ is f ample. Then, f is called *pl* flip contraction.
Definition : Let (X, Δ) be a pair such that $\lfloor \Delta \rfloor = S$ irreducible. The restricted algebra $R_S(K_X + \Delta)$ is given by

$$\bigoplus_{m \geq 0} Im(H^0(\mathcal{O}_X(m(K_X + \Delta))) \to H^0(\mathcal{O}_S(m(K_S + \Delta_S)))$$

Theorem

$f : X \to Z$ be a pl flipping contraction of a pair (X, Δ) such that $\lfloor \Delta \rfloor = S$. Then flip of f exists if and only if the restricted algebra is finitely generated.

Current status of MMP

- Around 1980, Mori proved the Cone and Contraction Theorem, a fundamental result in the Minimal Model Program in dim 3. Later Kawamata, Shokurov and others generalized the theorem in higher dimension.
- The idea of flip conjecture is also due to Reid. Lots of people , for example , Kawamata,Mori,Shokurov and others were trying to prove existence of flips.
- In 1987, Mori proved the existence of flips in dimension three, a breakthrough for which he was awarded the Fields Medal.

- In 2010, Christopher Hacon, James Mckernan, Caucher Birkar and Paolo Cascini proved **'flip exists in all dimensions'**.
- They could not prove termination conjecture in full generality. However,they proved **'termination of flips with scaling for varieties of general type'** which is enough to prove **'Existence of Minimal model program for varieties of general type'**.

Varieties of General Type

Definition

Let X be a projective variety.
We can define, rational maps $\phi_m : X \dashrightarrow \mathbb{P}(H^0(X, \mathcal{O}_X(mK_X))$.
Let Y_m be closure of image of ϕ_m.
Define $k(X) = \max \dim Y_m$.
If $H^0(X, \mathcal{O}_X(mK_X)) = 0 \forall m > 0$ then we say $k(X) = -\infty$.
If $k(X) = \dim X$ then variety is called a variety of **'general type'** and K_X is called **'big'**.

Main Results of [BCHM10]

- Existence of flips in all dimension.
- Termination of flips for scaling with an ample divisor for varieties of general type.
- For any smooth projective variety canonical ring is finitely generated.

Existence of flips

- The restricted algebra $R_S(K_X + \Delta)$ is defined as

$$\bigoplus_{m \geq 0} Im(H^0(\mathcal{O}_X(m(K_X + \Delta))) \to H^0(\mathcal{O}_S(m(K_S + \Delta_S)))$$

- If this restriction map is **surjective** then by **induction on dimension** the existence of flips follows.
- However, **in general**, the map is not surjective.
- In 2006, Hacon and Mckernan showed the restricted algebra can be identified with

$$\bigoplus_{m \geq 0} H^0(\mathcal{O}_S(m(K_S + \Theta))$$

  for some $0 \leq \Theta \leq \Delta_S$.
- This breakthrough allowed the use of induction on dimension to establish the existence of flips in all dimensions.

Let X be a normal projective variety with mild singularity. The Minimal Model Program(MMP) yields the following outcomes:

- If K_X is big then we can run MMP with scaling by an ample divisor and it terminates with a Minimal model.i.e $\phi : X \dashrightarrow Y$ such that K_Y is nef.
- If K_X is not pseudo-effective then we can run MMP with scaling by an ample divisor to get a Mori fiber space.

Proof of finite generation of canonical ring

- If $k(X) = -\infty$ then
  $H^0(X, \mathcal{O}_X(mK_X)) = 0 \forall m > 0 \implies R(K_X) = 0$
- If $0 \leq k(X) < n$ then we have a rational map $X \dashrightarrow Y \subseteq \mathbb{P}^N$.
  We replace X by $\tilde{X}$ and replace Y by $\tilde{Y}$, we get a morphism $f : \tilde{X} \to \tilde{Y}$ such that $K_{\tilde{X}} \sim_{\mathbb{Q}} f^*(K_{\tilde{Y}} + \theta)$.
  By canonical bundle formula, we get $\exists m, n \in \mathbb{N}$ such that

$$R(mK_{\tilde{X}}) \cong R(n(K_{\tilde{Y}} + \theta))$$

  Using induction of dimension we are done.

- If $k(X) = n$ then K_X is big. Then X has a minimal model, say Y.
- Let $\phi : X \dashrightarrow Y$,then $K_Y = \phi_* K_X$ is nef and big.
- Then by base point free theorem K_Y is semiample. i.e mK_Y is globally generated line bundle on Y for some $m > 0$.
  Then we can define a morphism $f : Y \to \mathbb{P}^k$ such that $mK_Y = f^*H$ where H is hyperplane divisor of $\mathbb{P}^k$. Then $R(mK_X)$ is finitely generated. $R(K_X)$ is finitely generated.

Thank You